



COMPUTER
ENGINEERING

QUEZON CITY

COURSE CODE/TITLE: CPE 009B

SECTION: CPE21S4

DATE PERFORMED: 1/08/2026

DATE SUBMITTED: 1/08/2026

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INSTRUCTOR: Engr. Jimlord Quejado

ACTIVITY NO. ____ [ACTIVITY NAME]

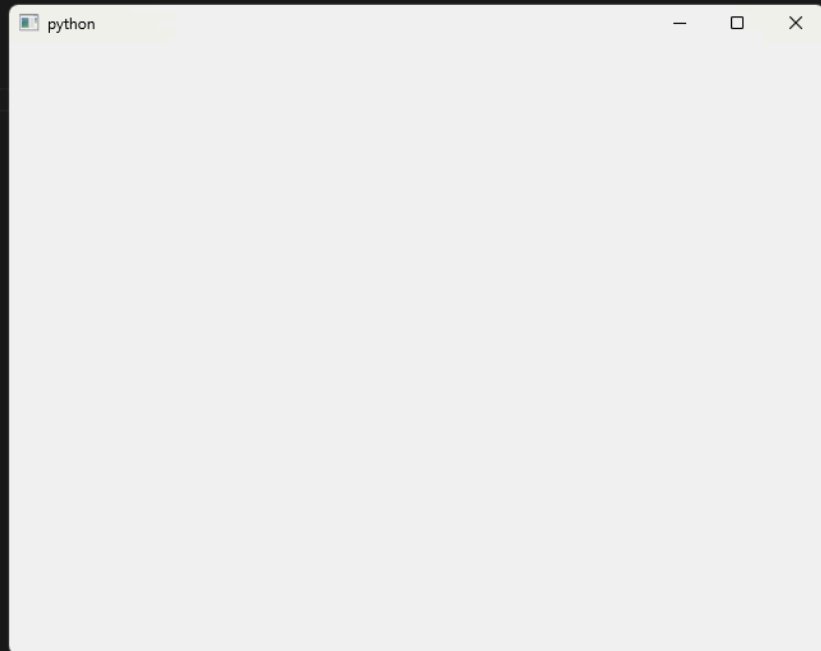
1. PROCEDURE OUTPUT

```
PS C:\Users\TIPQC\Documents\IGLOSO> python -m venv PyQt6
PS C:\Users\TIPQC\Documents\IGLOSO> pip install PyQt6
```

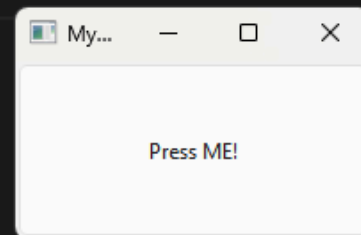
```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS
ion
+ FullyQualifiedErrorId : UnauthorizedAccess
PS C:\Users\TIPQC\Documents\IGLOSO> python -m venv PyQt6
PS C:\Users\TIPQC\Documents\IGLOSO> pip install PyQt6
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: PyQt6 in c:\users\tipqc\appdata\roaming\python\python312\site-packages (6.11.0)
Requirement already satisfied: PyQt6-sip<14,>=13.8 in c:\users\tipqc\appdata\roaming\python\python312\site-packages (from PyQt6) (13.11.1)
Requirement already satisfied: PyQt6-Qt6<6.12.0,>=6.11.0 in c:\users\tipqc\appdata\roaming\python\python312\site-packages (from PyQt6) (6.11.1)

[notice] A new release of pip is available: 25.0.1 -> 26.2
[notice] To update, run: python.exe -m pip install --upgrade pip
PS C:\Users\TIPQC\Documents\IGLOSO> ^C
PS C:\Users\TIPQC\Documents\IGLOSO> python.exe -m pip install --upgrade pip
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: pip in c:\users\tipqc\appdata\roaming\python\python312\site-packages (25.0.1)
Collecting pip
  Downloading pip-26.2-py3-none-any.whl.metadata (4.6 kB)
Downloading pip-26.2-py3-none-any.whl (1.8 MB)
1.8/1.8 MB 2.4 MB/s eta 0:00:00
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 25.0.1
    Uninstalling pip-25.0.1:
      Successfully uninstalled pip-25.0.1
  WARNING: The scripts pip.exe, pip3.12.exe and pip3.exe are installed in 'C:\Users\TIPQC\AppData\Roaming\Python\Python312\Scripts' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed pip-26.2
PS C:\Users\TIPQC\Documents\IGLOSO> 
```

```
CPE009B > HOA4P1 > main_app.py > ...
1 from PyQt6.QtWidgets import QApplication, QWidget
2
3 import sys
4
5
6 app = QApplication(sys.argv)
7 window = QWidget()
8 window.show()
9 sys.exit(app.exec())
```



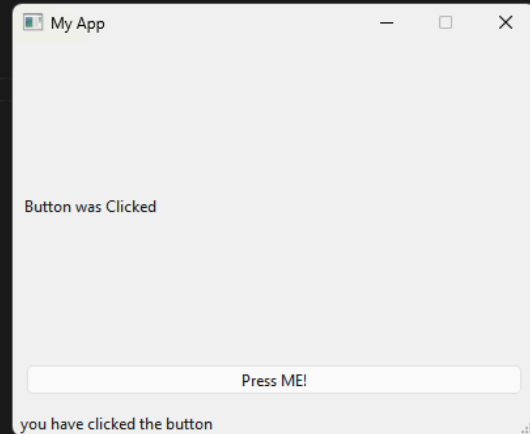
```
CPE009B > HOA4P1 > main_app.py > MainWindow > _init_
1 from PyQt6.QtWidgets import QApplication, QWidget, QPushButton, QMainWindow
2
3 import sys
4
5 class MainWindow(QMainWindow):
6     # constructor
7     def __init__(self):
8         super().__init__()
9
10        # Windows Properties
11        self.setWindowTitle("My App")
12        button = QPushButton("Press ME!")
13
14        self.setCentralWidget(button)
15
16
17 app = QApplication(sys.argv)
18 window = MainWindow()
19 window.show()
20 sys.exit(app.exec())
```



```

main_app.py 1 X
CPE009B > HOA4P1 > main_app.py > MainWindow > _init_
1  from PyQt6.QtWidgets import QApplication, QWidget, QLabel, QVBoxLayout, QPushButton, QMainWindow
2
3  import sys
4
5  class MainWindow(QMainWindow):
6      # constructor
7      def __init__(self):
8          super().__init__()
9
10         # Windows Properties
11         self.setWindowTitle("My App")
12         self.setFixedSize(400,300)
13
14         # widgets (child widgets)
15         self.label = QLabel("hello!")
16         self.button1 = QPushButton("Press ME!")
17
18         #arrange widgets vertically
19         layout = QVBoxLayout()
20         layout.addWidget(self.label)
21         layout.addWidget(self.button1)
22
23         #contain the widgets
24         container = QWidget()
25         container.setLayout(layout)
26         self.setCentralWidget(container)
27
28         #connect the button's clicked signal to the handler method(slot)
29         self.button1.clicked.connect(self.button_clicked)
30
31     def button_clicked(self):
32         self.label.setText("Button was Clicked")
33         self.statusBar().showMessage("you have clicked the button")
34
35 app = QApplication(sys.argv)
36 window = MainWindow()
37 window.show()
38 sys.exit(app.exec())

```



Conclusion/Observation

- In this activity, i learned the basics of coding with Pyqt. while following the instructions, i come up with a lots of error especially in the first code, the GUI is not showing because i cant run it . Eventually the problem is i need to download the code runner. After that i follow the code in the instructions and run it each. This last code above is the one with buttons, in order to have this, you need to import like QApplication, QWidget, QLabel, QVBoxLayout, QPushButton, QMainWindow, with this you can call it to have a function like the buttons.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Profile Card Code	Observation
<pre>profile_card.py X from PyQt5.QtWidgets import QApplication, QWidget, QLabel, QVBoxLayout, QPushButton, QMainWindow from PyQt5.QtCore import Qt import sys</pre>	I use spyder and PyQt5 because the vs code and PyQt6 doesnt work on my laptop so i use that instead.
<pre>6 class profileCard(QMainWindow): 7 def __init__(self): 8 super().__init__() 9 10 self.setWindowTitle("Profile Card") 11 self.setFixedSize(450,250)</pre>	This code is to set the window title "profile card" and the size of it to 450x250.
<pre>12 13 self.greetingLabel = QLabel("Who am I?") 14 self.greetingLabel.setAlignment(Qt.AlignCenter) 15 self.greetingLabel.setObjectName("greetingLabel") 16 self.button = QPushButton("Reveal") 17 # This makes the user know that the button is clickable 18 self.button.setCursor(Qt.PointingHandCursor)</pre>	I set the QLabel and align it at the center using the command Qt.AlignCenter, also i can call this by using from PyQt5.QtCore import Qt. and also i set the button of "reveal" and use the setCursor(Qt.PointinghandCursor) to tell the user that the button is clickable.
<pre>19 20 self.name_label = QLabel("") 21 self.name_label.setAlignment(Qt.AlignCenter) 22 self.name_label.setObjectName("nameLabel")</pre>	This part of code only make the nameLabel set its name and align it at the center.
<pre>23 24 self.status_label = QLabel("Click reveal to uncover my identity") 25 self.status_label.setAlignment(Qt.AlignCenter) 26 self.status_label.setObjectName("statusLabel")</pre>	Similar to the nameLabel, this is the same. This set its object name and align it at the center.
<pre>27 28 layout = QVBoxLayout() 29 layout.addWidget(self.greetingLabel) 30 layout.addWidget(self.button) 31 layout.addWidget(self.name_label) 32 layout.addWidget(self.status_label)</pre>	This part of code will make the layout of the contents of the QMainWindow. Its purpose is to position and organize the widgets from top to bottom.
<pre>33 34 container = QWidget() 35 container.setLayout(layout) 36 self.setCentralWidget(container) 37</pre>	This will hold the layouts.

<pre> 37 38 self.button.clicked.connect(self.show_profile) 39 self.apply_styles() 40 </pre>	This will be the destination where if you click it, you will be directed to another window and show the profile.
<pre> 40 41 def show_profile(self): 42 self.greetingLabel.setText("Hello! Denzen") 43 self.name_label.setText("Name: Denzen") 44 self.status_label.setText("Profile Load Successfully") 45 46 def apply_styles(self): </pre>	And this is another window after you clicked the button. This will be showing in the Window.
<pre> 45 46 def apply_styles(self): 47 self.setStyleSheet(""" 48 QMainWindow { 49 background-color: black;} 50 #greetingLabel { 51 color: white; 52 font-size: 20px; 53 font-weight: bold; 54 margin-bottom: 5px;} 55 #nameLabel { 56 color: violet; 57 font-size: 16px; 58 font-weight: bold; 59 margin-top: 5px;} 60 #statusLabel { 61 color: gray; 62 font-size: 11px; 63 font-style: italic; 64 margin-top: 5px;} 65 QPushButton { 66 background-color: blue; 67 color: white; 68 border: none; 69 border-radius: 4px; 70 padding: 10px 15px; 71 font-size: 13px;} 72 QPushButton:hover { 73 background-color: #007399;} 74 """) </pre>	This is where the styles of the fonts, backgrounds, colors etc.. this is where you manipulate all of the text and colors in order to show the program neatly and organize.
<pre> 74 75 if __name__ == "__main__": 76 app = QApplication.instance() or QApplication(sys.argv) 77 window = profileCard() 78 window.show() 79 sys.exit(app.exec()) </pre>	After you run the code this is the part of the code that will make the window show or pop on your screen.



Supplementary Questions:

1. How does the layout affect how your app looks and behaves when resized?
 - The design automatically scales and repositions the widgets in the window to keep the layout neat and proportionate while stretching. If I resize it 500 by 250, then the window will stretch along with all that's in it like fonts, buttons, colors, margins etc.
2. What would happen if you tried to update a widget that wasn't added to the layout?
 - The data will still be modified through the code, but it will not be shown in the form of the widget within the application.
3. How would you expand this app to include input fields for the name instead of hardcoding it?
 - I will place a QLineEdit widget in the layout and call its .text() method in the show_profile() function.

Conclusions/Learnings:

- Through this exercise, I got to learn how to build a working GUI program using PyQt5 widgets such as QLabel, QPushButton, and QVBoxLayout. It allowed me to gain practical knowledge on how to bind button clicks to specific slot functions to update the text on the window in real time. In addition, the use of PyQt style sheets helped me learn how CSS properties can turn an ordinary window into an app through customization.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: google

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: Code assistances, grammar correction

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: I use it to search for the right PyQt imports for specific things like this Qt.AlignCenter. I also use it for grammar correction

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or



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completion of the required work.

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I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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